

Yiming Bao

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EDUCATION

- **Student Intern**

Kempner Institute & School of Engineering and Applied Sciences, Harvard University
(advisors: Heng Yang, Qianqian Wang, Yilun Du)

Jun 2026 - Present
Cambridge, MA, USA

- **Final-Year Undergraduate**

Department of Computer Science and Technology & Xinya College, Tsinghua University
(advisor: Huazhe Xu)
GPA: 3.8/4.0

2023 - Present
Beijing, China

- **Selected Major Courses**

CS: Programming and Training (4.0), Data Structure (4.0), Software Engineering (4.0), High Performance Computing (4.0), Digital Logic Circuit (4.0), Media Computing (4.0), Computer Network (4.0), Cybersecurity Fundamentals (4.0)
AI/ML: Introduction to Artificial Intelligence, Artificial Neural Network (4.0), LLMs-Driven Software Development (4.0)
Math/Physics: Probability and Statistics (4.0), Linear Algebra (4.0), Calculus (4.0), Complex Analysis (4.0), Numerical Analysis (4.0), Physics(I)(III) (4.0)

INTERNSHIPS

- **Research Lead**, Stealth Startup
Cambridge, MA, USA

Jul 2026 - Present

- **Research Intern**, ByteDance Seed-Robotics, Robotics Large Model Group
Beijing, China

Apr 2026 - Jul 2026

PUBLICATIONS & PREPRINTS

- **Verifiable 4D World-Action Model for Robust Robot Control**

ICLR 2027 (In Submission) [Project]

- **UniDex: A Robot Foundation Suite for Universal Dexterous Hand Control from Egocentric Human Videos**

CVPR 2026 [Project][Paper] [Code] [Model] [X]

Gu Zhang*, Qicheng Xu*, [...], Yiming Bao*, [...], Yang Gao[†], Hao Zhao[†], Hang Zhao[†], Huazhe Xu[†] (19 authors)

- **DexImit: Learning Bimanual Dexterous Manipulation from Monocular Human Videos**

RSS 2026 [Project] [Paper] [Code] [X]

Juncheng Mu*, Sizhe Yang*, Yiming Bao, Hojin Bae, Tianming Wei, Linning Xu, Boyi Li[†], Huazhe Xu[†], Jiangmiao Pang[†]

- **SiMDex: Mining Similar Egocentric Videos for Cross-Embodiment Dexterous Manipulation**

ICLR 2027 (In Submission) [Project] [Paper] [Video]

Nie Lin, Takehiko Ohkawa, [...], Yiming Bao, [...], Xiao Ma, Wei Xu, Yoichi Sato (13 authors)

RESEARCH

- **Research Interest**

My long-term research goal is to develop robots with human-level versatility in **perception, reasoning, and manipulation**, ultimately enhancing everyday life at scale.

My current work emphasizes **dexterous manipulation**, leveraging **human data** to address the representation gap and data scarcity that limit multi-finger manipulation systems. I pursue this through two complementary directions:

- (i) **real-to-sim & sim-to-real transfer** — reconstructing human demonstrations in simulation to generate scalable, physically grounded robot training data;
- (ii) **direct learning from human data** — using large-scale human hand–object interaction data to close the embodiment and representation gap for dexterous manipulation policies.

• Experience

- **S4NDBOX: Verifiable 4D World-Action Model** Jul 2026 – Sep 2026
 - * Developed a **spatial-forcing 4D world-action model** that jointly predicts RGB, metric depth, geometric motion fields, and robot actions; introduced geometric consistency scoring for **verification-guided candidate selection**
 - * Improved **RoboTwin Hard success by 13.3 points** over the RGB-only baseline; in zero-shot sim-to-real Stack Cups, improved mean task score by **41.7 points (three-view)** and **40.0 points (single-view)** under real-world shifts
- **SiMDex: Mining Similar Egocentric Videos for Cross-Embodiment Dexterous Manipulation** Feb 2026 – Aug 2026
 - * Try to answer **which human data actually benefits dexterous manipulation**, casting human data selection for **VLA post-training** as a **recommendation problem**
 - * Build a **three-layer recall–ranking–re-ranking pipeline** that mines task-relevant subsets from a pool of **~32M egocentric human samples**, operating in a **morphology-agnostic action space** requiring no change to VLA architecture or training
 - * Using only **~1.49M mined samples (<5% of the pool)**, improve the overall success rate from **47.7% to 61.1%** over an equal-budget random-sampling baseline, showing selective curation beats indiscriminate data mixing
- **DexImit: Learning Bimanual Dexterous Manipulation from Monocular Human Videos** Nov 2025 - Jan 2026
 - * Try to deal with **data scarcity** that limits the generalizations of bimanual dexterous manipulation using **generated or in-the-wild human videos**
 - * A simple and effective framework automatically converts **monocular human videos** into **physically plausible robot demonstrations** via 4D reconstruction, action-centric scheduling, force-closure grasp synthesis, and large-scale augmentation
 - * enabling zero-shot real-world deployment **without any robot data**
- **UniDex: A Robot Foundation Suite for Universal Dexterous Hand Control from Egocentric Human Videos** Jun 2025 – Nov 2025
 - * Try to address **data scarcity and embodiment heterogeneity** in dexterous manipulation by leveraging large-scale **ego-centric human videos** to build a robot-centric foundation dataset
 - * Construct **UniDex-Dataset** (10M frames, 50K trajectories, 8 hands) and propose a unified **Function–Actuator–Aligned Space (FAAS)** with a pretrained **3D VLA policy** for cross-hand dexterous control
 - * Enabling strong **zero-shot spatial, object, and cross-hand generalization**, significantly outperforming prior VLA baselines in real-world tool-use tasks
- **Co-training Egocentric Human Data with Simulation and Teleoperation Robot Data** Apr 2026 -
 - * Investigating how to best **co-train on egocentric human data together with simulation and teleoperation robot data**, and what mixing strategies transfer across the embodiment gap
- **Tactile World Model for Test-Time Planning** Jul 2026 -
 - * Building a **tactile-conditioned world model** that predicts contact-rich interaction outcomes, enabling **test-time planning** for dexterous manipulation

SKILLS

- **Language:** TOEFL (110/120, 5.5/6.0), CET-4, CET-6
- **Programming and Scripting Languages:** Python, C/C+, Rust, Java, JavaScript/TypeScript, CUDA, SystemVerilog
- **Professional Software:** Pytorch, ROS, Sapien, IsaacLab, MuJoCo, Polymetis
- **Hardware and Robots:** Franka Panda, OpenArm, Inspire Hand, Wuji Hand, X Hand, Sharpa Wave Hand, Robotiq Gripper, Manus Glove, OpenTouch Glove, Apple Vision Pro, Oculus Quest, Realsense Camera, NUC

MISC

- **Campus Activities:** Member of the Tsinghua University Body Building Team, Tsinghua University Dance Team, Xinya College Basketball Team, and the Student Union of Xinya College
- **Before College:** Quite into Physics in high school; participated in the Physics Olympiad
- **Hobbies:** Body building, basketball, tennis, hiking, guitar, and Chinese rock and folk music